PATRICK BARRY
Los Angeles | pbarry670@gmail.com
https://github.com/pbarry670
https://www.linkedin.com/in/pbarry670
U.S. citizen. Active Top Secret clearance.

EDUCATION

M.S. Aerospace Engineering, Georgia Institute of Technology - May 2026
Completed, through the BS/MS honors program. Bachelor's and master's both at Georgia Tech.
B.S. Aerospace Engineering, Georgia Institute of Technology - May 2025
Finished in three years. Final GPA 4.0.
Relevant coursework: Optimal Guidance & Control, Control System Design/Analysis, Robotics & Autonomy, Orbital Mechanics, Optimization-Based Learning & Control, Optical Navigation.

EXPERIENCE

GNC Engineer - SpaceX
Starshield | June 2026 - Present
- I work on beam planning and tracking algorithms.

Guidance, Navigation, and Control Intern - Johns Hopkins University Applied Physics Laboratory
Laurel, MD | May 2025 - August 2025
- Designed error-state Kalman filters and robust flight control systems.
- Built a MATLAB/Simulink toolset to compute stability margins and performance metrics across launch-vehicle trajectories under model uncertainty.
- Implemented a Kalman filter for post-flight error analysis of an inertially guided launch vehicle, from an IMU error model and measurements along the trajectory.
- Designed and ran experiments to estimate error coefficients of a navigation-grade accelerometer, including high-g vibration-table and centrifuge testing.
- Processed that data with digital filters and Fourier transforms, and estimated the coefficients with an unscented Kalman filter.

Undergraduate Research Assistant, Autonomous Racecar Group - Dynamics and Control Systems Laboratory, Georgia Tech
Atlanta, GA | January 2024 - December 2024
- Wrote a Model-Predictive Path Integral controller for the lab's F1Tenth racecars.
- Implemented it in Python with PyTorch for GPU acceleration, and tied it into the car through ROS.
- Brought the loop from a few hertz up to 40 Hz. My resume summarizes that as a 13x increase, from 3 Hz to 40 Hz. The racecar page has the longer test notes: about 4 Hz at first, 21 Hz after a custom PyTorch extension, then 40 Hz.
- Wrote a visualization so I could debug and tune the cost without a full pad setup, and tested at the Indoor Flight Lab.

Vehicle Design and Analysis Intern - The Aerospace Corporation
Chantilly, VA | May 2024 - August 2024
- Modeled spacecraft concept designs in a concurrent engineering tool against mission architectures.
- Worked with subject-matter experts on whether existing designs were feasible, and went through satellite PDRs and CDRs to see how changes moved size, mass, and power.
- Analyzed attitude-control and propulsion subsystems for performance against cost, then presented the study and wrote up the documentation.

Avionics & Controls Lead - Ramblin' Rocket Club, GNC project

Georgia Tech | August 2023 - May 2025
- Led a subteam of 25+ students on avionics, software, and controllers for rockets steered by jet vanes and by gimbaling the motor.
- Linearized a nonlinear rocket model into trim points and a state-space model along the flight.
- Led GNCSIM, an object-oriented MATLAB 6-DOF simulation with LQR about a reference trajectory, plus Monte Carlo under parameter uncertainty.
- Wrote position, velocity, and attitude estimation in C on an STM32.
- Built the static-fire avionics: remote command, motor ignition, and servo timing.

Undergraduate Research Assistant, Sensor Development - Aerospace Systems Design Laboratory, Georgia Tech
Atlanta, GA | August 2023 - December 2023
- Worked on a small, modular, low-cost current sensor for building energy use.
- Designed the PCB in KiCad and the housing in SolidWorks. A full unit came to $102.
- The resume also notes current-sensor software on a Raspberry Pi. The longer build notes are on the project page.

Deputy Project Manager - NASA L'SPACE Mission Concept Academy
Virtual | January 2023 - May 2023
- Deputy project manager on an 11-person team writing a Preliminary Design Review for a rover mission to Mars' northern polar region.
- Ran communication between the science and engineering subteams on timeline, operations, prior missions, and budget.

Tennis Instructor - Mitchell Frank Tennis Academy
Annandale, VA | Summers 2021, 2022, and 2023
- Ran drills and games for kids ages 4-14.
- Promoted to manager of the primary location for summer 2023, overseeing multiple coaches and more than 50 students.
- Directed the summer 2023 fitness program for players working toward college recruitment.

Wilderness Outpost Attendant - Campus Recreation Center, Georgia Tech
Atlanta, GA | January 2023 - May 2023
- Prepped outdoor gear for trips, staffed the desk, and registered people for outings.

PUBLICATIONS
Development of an LQR Controller for a Jet Vanes Thrust-Vectored Rocket. Barry, Zheng, Caillet, and Reynolds. AIAA Regional Student Conferences, 2025.
https://arc.aiaa.org/doi/10.2514/6.2025-99074
Integrated State Estimation Development for Actively Controlled Rockets. Zheng, Barry, Ambati, and Vadia. AIAA Regional Student Conferences, 2025.
https://arc.aiaa.org/doi/10.2514/6.2025-99521

PROJECTS
Jet vanes rocket. State estimation and an LQR flight controller for a rocket steered by jet vanes in the motor plume. I led the avionics, the C flight code, and GNCSIM.
https://github.com/pbarry670/GNCSIM
Autonomous racecar. MPPI for Georgia Tech's F1Tenth cars. The controller does not need the track in advance. It builds a centerline from LiDAR and picks a lowest-cost rollout.
https://github.com/pbarry670/f1tenth-mppi
Ballistic missile tracking. Infrared tracking of a ballistic missile from a satellite constellation, with an extended Kalman filter that starts from one line of sight or from a

triangulation. https://github.com/pbarry670/BallisticMissileTracking

Planetary supply drop. A C++ simulation of dropping a payload from orbit onto a chosen point on Earth: orbit control, ballistic entry, a parachute, and a fuel-optimal landing. https://github.com/pbarry670/Planetary-Supply-Drop

Drone trajectory optimization. Differential dynamic programming for a quadrotor, with barrier costs around obstacles, extended to receding-horizon MPC. https://github.com/pbarry670/drone-ddp

Thrust-vectored rockets. Gru and Vector, two small rockets whose motors gimbal. My first project as avionics and controls lead: a three-month build, with new members coming on while we were still designing.

Low-cost current transformer. A small current sensor for building managers who want appliance-level power. PCB in KiCad, housing in SolidWorks, about $102 a unit, with the readings sent out over the network.

QuadGNC. In-house GNC, simulation, and flight software for a quadcopter. The public repo is the collection itself. https://github.com/pbarry670/QuadGNC